Assignment: Healthcare Risk Prediction — Machine Learning vs Neural Networks

1. **Problem Understanding**

Description of the Problem

The goal of this project is to predict whether a patient has diabetes using basic health information such as age, BMI, exercise level, smoking status, and blood pressure. The prediction result is binary:

0 – No Diabetes

1 – Diabetes

This helps identify patients who may be at risk.

2. **Importance of Predicting Diabetes Risk**

1. Helps detect diabetes early
2. Allows patients to take preventive actions
3. Reduces health complications and medical costs
4. Supports better healthcare decision-making

3. Explorartory Data Analysis

1. The total number of records in the datasets is **6000.**
2. No of non smokers are : **4158**
3. No of smokers are : **1842**
4. No of smokers who have diabetes are: **161**
5. No of non-smokers who are diabetic are :**381**
6. Average age of smokers is : **51.51**
7. Average Blood Pressure of smokers is : **141.41**
8. Average Blood Pressure of non-smokers is: **131.04**
9. Majority of the patients have age **32 .**
10. From the box plot we can see that median age of diabetic patients is above **60**.
11. From the correlation heatmap we can find that
    1. Age and blood pressure are highly positively correlated that means age increases blood pressure also increases
    2. Also, smoker, bmi, blood pressure has some amount of impact on diabetes .

4 . Models Explanation

1. For machine learning model the logistic regression model is trained.
2. The data is split into **80%** training data and **20%** testing data.

Logistic Regression is a machine learning algorithm used for classification problems, mainly when the output has two possible outcomes, such as: Yes / No, 0 / 1,Diabetes / No Diabetes. Logistic Regression looks at input features (like age, BMI, blood pressure, etc.) and predicts the

probability that an outcome belongs to a certain class. Instead of predicting direct value, it predicts a number between 0 and 1 using a special function called the sigmoid function:

If the probability ≥ 0.5 → Class 1

If the probability < 0.5 → Class 0

Simple and easy to understand

Fast and efficient

Works well for binary classification problems

Widely used in medical predictions like diabetes, heart disease, and cancer detection

Example in Diabetes Prediction

Using inputs such as: Age, BMI, Exercise hours, Smoking status, Blood pressure

Logistic Regression predicts whether a patient has diabetes (1) or does not have diabetes (0).

Neural Network Architecture

A Sequential Neural Network is a type of neural network where layers are arranged one after another in a straight line. Each layer passes its output directly to the next layer.

Input Layer

Takes input features (such as age, BMI, blood pressure, etc.).

It does not perform calculations, only passes data forward.

Hidden Layer(s)

These layers perform calculations using weights and activation functions like ReLU.

They learn patterns and relationships in the data.

Output Layer

Produces the final prediction.

For binary classification (like diabetes prediction), it usually has one neuron with a sigmoid activation function, giving output as 0 or 1.

An activation function decides whether a neuron should be active or not.

It helps a neural network learn patterns by adding non-linearity (so the model can understand complex relationships).

In the Neural Network trained I have taken 32 inputs neurons (1 input layer), 16 hidden neurons(1 hidden layer) and 1 output neuron(output layer).

For neural network also the training data is 80% and testing data is 20%.

5 . Model Evaluation

Machine Learning Model(Logistic Regression)

Accuracy is 0.9092

Precision: 0.00

Recall : 0.00

F1_score : 0.00

Confusion matrix is

[[1091 0]

[109 0]]

Neural Network matrix is

[[1090 1]

[109 0]]

Accuracy is 0.9092

Precision: 0.00

Recall : 0.00

F1_score : 0.00

There is a slight difference between the confusion matrix of both ml model and dl model.

Comparative Analysis

The Logistic Regression model and the Neural Network model show very similar performance behavior:

Accuracy for both models is 0.9092, which appears high at first glance.

Both models failed to correctly predict any positive class instances, as indicated by:

Precision = 0.00

Recall = 0.00

F1-score = 0.00

Logistic Regression predicts all samples as the negative class.

Neural Network predicts only one positive sample, which is still incorrect (false positive).

Neither model identifies any true positives.

Key Difference:

Logistic Regression is **fully biased toward the majority class**.

Neural Network shows slightly more flexibility but still fails to learn meaningful patterns for the minority class.

Reasons for Accuracy Differences:

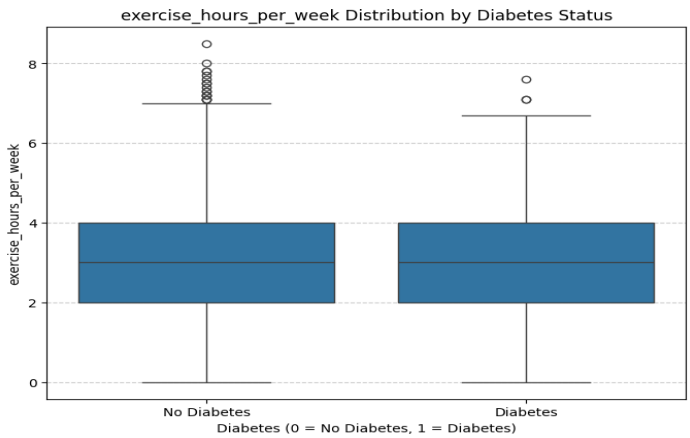
1. Severe Class Imbalance

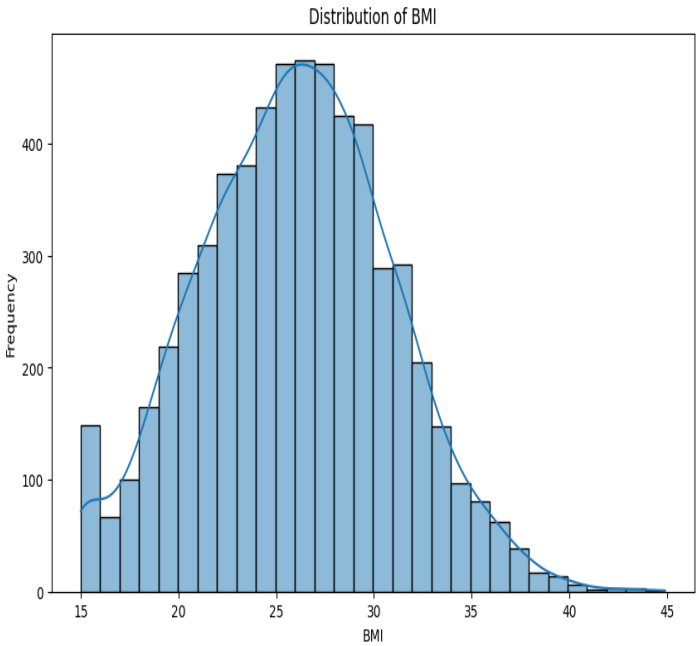
1. The dataset is heavily skewed toward the negative class.
2. Predicting all samples as negative yields high accuracy by default.

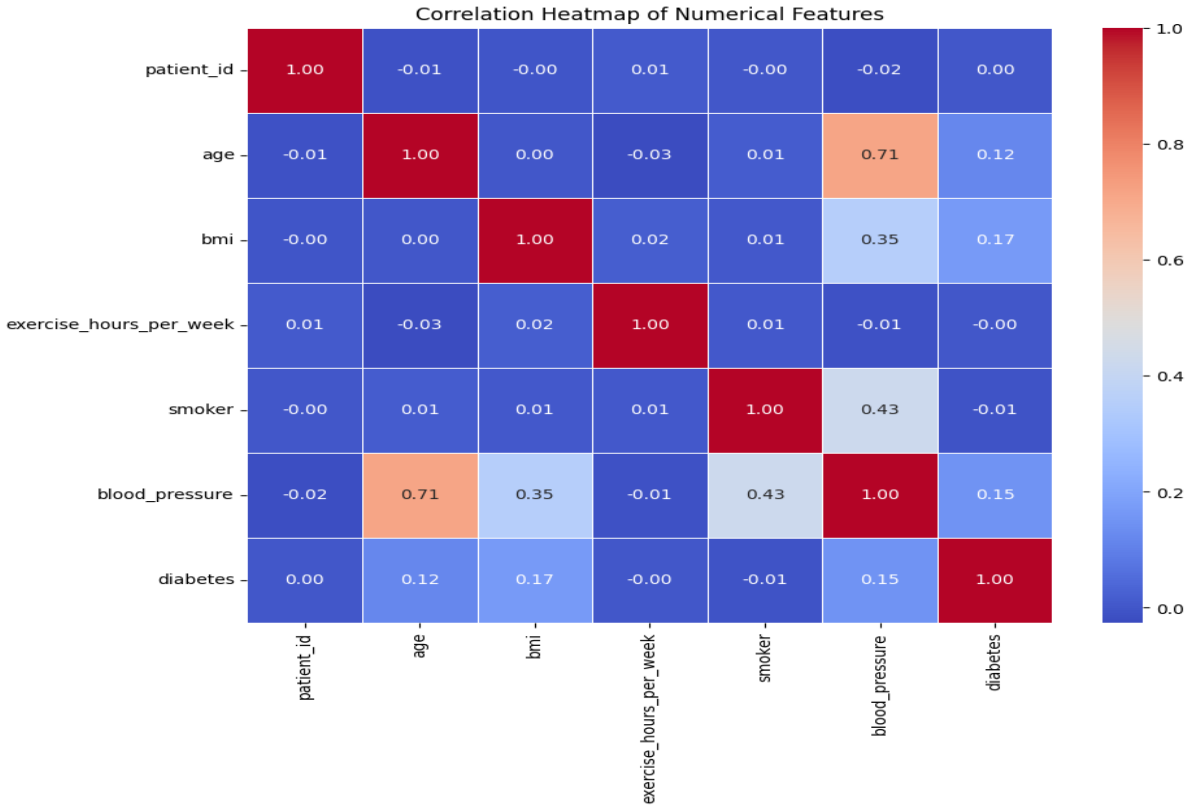
2. Accuracy Is a Misleading Metric
   1. Accuracy reflects correct overall predictions, not class-wise performance.
   2. Since the majority class dominates, models appear “accurate” even when they fail completely on the minority class.

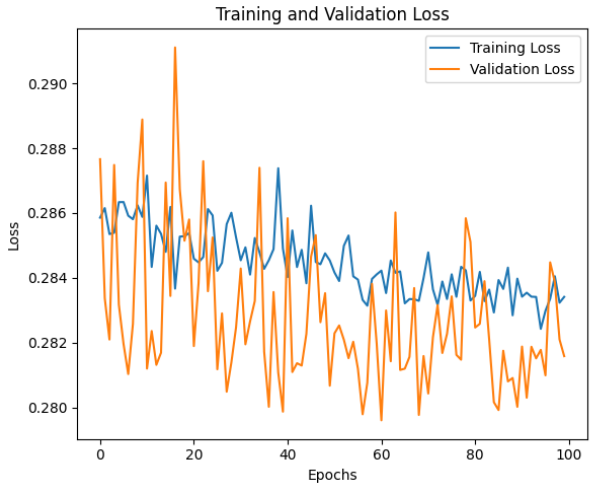
Recommendations for real world use cases

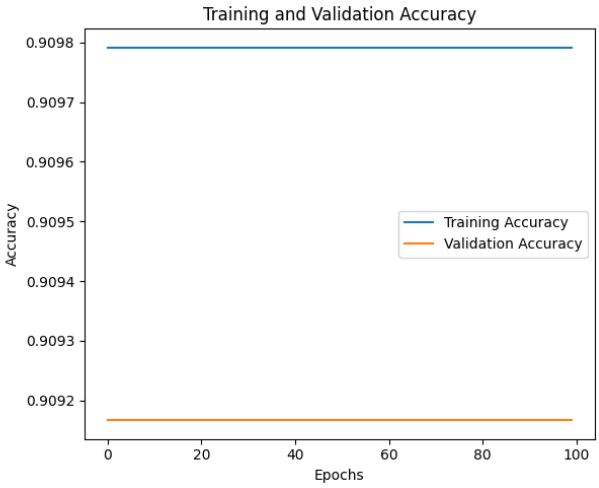
- Handle Class Imbalance
- Use Better Evaluation Metrics
- Threshold Optimization
- Feature Engineering
- Model Selection

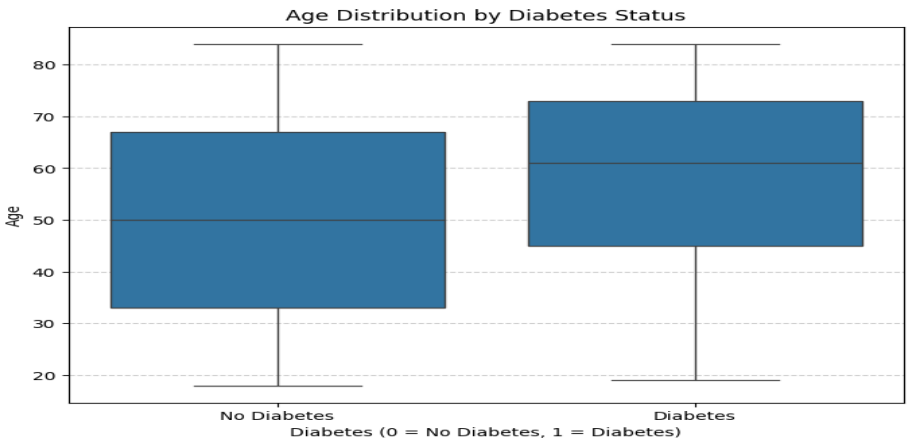





Correlation Heatmap of Numerical Features
patient_id
age
bmi
exercise_hours_per_week
smoker
blood_pressure
diabetes
1.00 -0.01 -0.00 0.01 -0.00 -0.02 0.00
-0.01 1.00 0.00 -0.03 0.01 0.71 0.12
-0.00 0.00 1.00 0.02 0.01 0.35 0.17
0.01 -0.03 0.02 1.00 0.01 -0.01 -0.00
-0.00 0.01 0.01 0.01 1.00 0.43 -0.01
-0.02 0.71 0.35 -0.01 0.43 1.00 0.15
0.00 0.12 0.17 -0.00 -0.01 0.15 1.00
1.0
0.8
0.6
0.4
0.2
0.0


Training and Validation Loss
Training Loss
Validation Loss
Loss
Epochs
0.290
0.288
0.286
0.284
0.282
0.280
0
20
40
60
80
100


Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.9098
0.9097
0.9096
0.9095
0.9094
0.9093
0.9092
0
20
40
60
80
100


Age Distribution by Diabetes Status
Age
80
70
60
50
40
30
20
No Diabetes
Diabetes
Diabetes (0 = No Diabetes, 1 = Diabetes)